

A First-Year-University Explainer For Standard-Normal GHQ, SGQF Levels, Tensor Labels, Sparse-Grid Level, And Smolyak Construction

Chak Wong

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Abstract

This note is written for a reader who keeps seeing phrases such as “level-2 rule,” “tensor rule $(2, 2)$,” “sparse-grid level $L = 2$,” and “Smolyak construction,” but does not yet feel that those phrases mean something concrete. The goal is not to present new theory. The goal is to explain, slowly and with explicit examples, what those phrases mean in the sparse-grid quadrature filtering material based on Jia, Xin, and Cheng. The central message is simple: the same word “level” is being used for different jobs. A one-dimensional rule level labels a rule family member such as I_1 or I_2 . A tensor label such as $(2, 2)$ says which one-dimensional level is used in each coordinate. A sparse-grid level such as $L = 2$ says which tensor rules are allowed into the sparse-grid approximation. Smolyak construction means: take several tensor rules, attach plus or minus coefficients to them, and merge repeated point contributions. Those are related ideas, but they are not the same idea.

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1 If you remember only six sentences

If the rest of the note feels long, keep these six sentences in mind.

1. A rule such as I_1 , I_2 , or I_3 is a *one-dimensional* weighted-average recipe for approximating a Gaussian integral.
2. In BayesFilter’s default convention, these one-dimensional rules come from standard-normal Gauss–Hermite quadrature, abbreviated standard-normal GHQ.
3. A label such as $(2, 1)$ or $(2, 2, 2)$ says: use one rule level in each coordinate, then take the Cartesian product.
4. A sparse-grid level such as $L = 2$ does *not* mean the same thing as the tensor label $(2, 2)$ or $(2, 2, 2)$.
5. Smolyak construction means: take several tensor rules, attach plus or minus coefficients to them, add them together, and merge repeated points.
6. The numbers 1, 2, 3 in these labels are not polynomial degrees. They are level labels, and the source paper ties level ℓ to exactness of at least degree $2\ell - 1$ in one dimension [1].

2 What are we trying to approximate?

Before discussing sparse grids, it helps to remember the original problem. We want to approximate a Gaussian integral such as

$$\int_{\mathbb{R}} \psi(\xi) \phi(\xi) \, d\xi, \quad \phi(\xi) = \frac{1}{\sqrt{2\pi}} e^{-\xi^2/2}. \quad (1)$$

The function ϕ is the standard-normal density. The function ψ is whatever function we want to average.

A quadrature rule replaces the integral by a weighted sum:

$$I_\ell[\psi] = \sum_{a=1}^{s_\ell} \omega_{\ell,a} \psi(\zeta_{\ell,a}). \quad (2)$$

Here:

- $\zeta_{\ell,a}$ are the nodes, meaning the places where we evaluate ψ ,
- $\omega_{\ell,a}$ are the weights, meaning how much each evaluation counts,
- s_ℓ is the number of nodes used by level ℓ .

So a one-dimensional rule level is simply a particular weighted-average recipe.

3 What is standard-normal GHQ?

This is one of the things that should be stated more explicitly.

3.1 Plain-English meaning

GHQ stands for *Gauss–Hermite quadrature*. In the BayesFilter chapter, the default family of one-dimensional rules is the standard-normal GHQ family.

You do *not* need to know Hermite polynomials to read the note. For our purposes, you only need to know this:

standard-normal GHQ means a precomputed list of nodes and weights designed to approximate integrals against the standard-normal density $\phi(\xi)$.

3.2 BayesFilter default family

The BayesFilter chapter fixes the default family as

$$I_\ell = \text{ONEDIMGHQ}(2\ell - 1). \quad (3)$$

This means:

- level 1 uses a 1-point standard-normal GHQ rule,
- level 2 uses a 3-point standard-normal GHQ rule,
- level 3 uses a 5-point standard-normal GHQ rule,
- in general, level ℓ uses a $(2\ell - 1)$ -point standard-normal GHQ rule.

So in BayesFilter the family is not mysterious:

$$I_1, I_2, I_3, \dots \quad \text{means} \quad 1\text{-point, } 3\text{-point, } 5\text{-point, } \dots \text{standard-normal GHQ rules.} \quad (4)$$

3.3 Concrete low-level members

The first two members are the ones used repeatedly in the teaching note.

Level 1. The level-1 rule is

$$I_1 : (0, 1). \quad (5)$$

This means: evaluate only at the center point 0 and give it full weight 1. So

$$I_1[\psi] = \psi(0). \quad (6)$$

Level 2. The level-2 rule is the familiar 3-point standard-normal GHQ rule

$$I_2 : \left\{ \left(-\sqrt{3}, \frac{1}{6} \right), \left(0, \frac{2}{3} \right), \left(\sqrt{3}, \frac{1}{6} \right) \right\}. \quad (7)$$

This means

$$I_2[\psi] = \frac{1}{6}\psi(-\sqrt{3}) + \frac{2}{3}\psi(0) + \frac{1}{6}\psi(\sqrt{3}). \quad (8)$$

Level 3. Level 3 means: use the 5-point standard-normal GHQ rule. The original teaching note does not write out those five nodes and weights explicitly, but under the BayesFilter default policy level 3 means “the 5-point standard-normal GHQ member of the same family.”

3.4 Why the level number is not a polynomial degree

The label “2” in I_2 is a family label, not the degree of a polynomial. For the concrete rule (7), one can check

$$I_2[1] = 1, \tag{9}$$

$$I_2[\xi^2] = \frac{1}{6}(3) + \frac{2}{3}(0) + \frac{1}{6}(3) = 1, \tag{10}$$

$$I_2[\xi^4] = \frac{1}{6}(9) + \frac{2}{3}(0) + \frac{1}{6}(9) = 3. \tag{11}$$

Because the rule is symmetric, odd powers cancel automatically:

$$I_2[\xi] = I_2[\xi^3] = I_2[\xi^5] = 0. \tag{12}$$

So the specific 3-point rule called I_2 is exact through degree 5 under the standard-normal weight. That is why “level 2” cannot mean “degree 2.”

4 What does the function symbol mean?

Another thing that easily feels too abstract is the symbol F .

4.1 Plain-English meaning

When we write

$$I_i[F] \quad \text{or} \quad A_{b,L}[F], \tag{13}$$

we mean:

apply the quadrature rule to the function F .

So F is just the function we want to average under the Gaussian weight.

In one dimension I often wrote ψ . In several dimensions I switch to F . They play the same role.

4.2 Concrete example used later

To make everything concrete, later we will use the three-variable function

$$F(\xi_1, \xi_2, \xi_3) = e^{\xi_1 \xi_2 \xi_3}. \tag{14}$$

This is a good teaching example because:

- it is clearly nonlinear,
- it depends on all three variables together,
- if any one coordinate is 0, then the product $\xi_1 \xi_2 \xi_3$ becomes 0 and $F = 1$,
- but at the full three-way corner points $(\pm\sqrt{3}, \pm\sqrt{3}, \pm\sqrt{3})$ the function is not 1.

That makes it ideal for seeing what a low sparse-grid level can and cannot see.

5 The three jobs done by the word “level”

This section is the heart of the note. It separates the three jobs as clearly as possible.

5.1 Job 1: level of a one-dimensional rule

The notation I_ℓ means:

“the one-dimensional quadrature rule at level ℓ .”

This is a one-dimensional object.

5.2 Job 2: level inside a tensor label

The notation

$$\mathbf{i} = (i_1, \dots, i_b)$$

means:

“in coordinate 1 use level i_1 , in coordinate 2 use level i_2 , and so on.”

So a tensor label such as $(2, 1)$ is not a new kind of level. It is just a short way to list which one-dimensional rule level is used in each coordinate.

5.3 Job 3: sparse-grid level

The notation L in the sparse-grid rule means something different again. It controls which tensor rules are allowed into the sparse-grid approximation.

So when you see

$$L = 2,$$

you should hear:

“use the sparse-grid combination rule at level 2.”

You should *not* hear:

“keep the full tensor rule $(2, 2)$ ” or “keep the full tensor rule $(2, 2, 2)$.”

5.4 One-line dictionary

Here is the shortest useful dictionary:

$$\begin{aligned} I_2 &: \text{one-dimensional rule,} \\ (2, 2) &: \text{two-dimensional tensor label,} \\ (2, 2, 2) &: \text{three-dimensional tensor label,} \\ L = 2 &: \text{sparse-grid combination level.} \end{aligned}$$

These symbols are related, but they are not interchangeable.

6 Tensor labels explained slowly

Suppose we move from one variable ξ to two variables (ξ_1, ξ_2) . Now we can choose one one-dimensional rule for the first coordinate and one for the second coordinate.

The multi-index

$$\mathbf{i} = (i_1, i_2)$$

records those choices.

6.1 What $(1, 1)$ means

The label $(1, 1)$ means:

use I_1 in coordinate 1 and I_1 in coordinate 2.

Because each I_1 has only one point, the tensor product has only one point:

$$(0, 0) \quad \text{with weight} \quad 1. \quad (15)$$

So for any two-variable function F ,

$$I_{(1,1)}[F] = F(0, 0). \quad (16)$$

6.2 What $(1, 2)$ means

The label $(1, 2)$ means:

use I_1 in coordinate 1 and I_2 in coordinate 2.

So the first coordinate stays fixed at 0, while the second coordinate uses the three points of I_2 . Therefore the tensor points are

$$(0, -\sqrt{3}), \quad (0, 0), \quad (0, \sqrt{3}), \quad (17)$$

with weights

$$\frac{1}{6}, \quad \frac{2}{3}, \quad \frac{1}{6}. \quad (18)$$

So

$$I_{(1,2)}[F] = \frac{1}{6}F(0, -\sqrt{3}) + \frac{2}{3}F(0, 0) + \frac{1}{6}F(0, \sqrt{3}). \quad (19)$$

6.3 What $(2, 1)$ means

The label $(2, 1)$ means:

use I_2 in coordinate 1 and I_1 in coordinate 2.

So now the second coordinate stays fixed at 0, while the first coordinate uses the three points of I_2 . Therefore the tensor points are

$$(-\sqrt{3}, 0), \quad (0, 0), \quad (\sqrt{3}, 0), \quad (20)$$

with the same weights

$$\frac{1}{6}, \quad \frac{2}{3}, \quad \frac{1}{6}. \quad (21)$$

6.4 What $(2, 2)$ means

The label $(2, 2)$ means:

use I_2 in coordinate 1 and I_2 in coordinate 2.

Because each coordinate contributes three points, the full tensor product has

$$3 \times 3 = 9 \tag{22}$$

points:

$$\begin{aligned} &(-\sqrt{3}, -\sqrt{3}), (-\sqrt{3}, 0), (-\sqrt{3}, \sqrt{3}), \\ &(0, -\sqrt{3}), (0, 0), (0, \sqrt{3}), \\ &(\sqrt{3}, -\sqrt{3}), (\sqrt{3}, 0), (\sqrt{3}, \sqrt{3}). \end{aligned} \tag{23}$$

Very important warning. The tensor label $(2, 2)$ does *not* mean sparse-grid level 2. It only means “use I_2 in both coordinates.”

7 The local index α : which actual point did you choose?

After choosing the tensor label $\mathbf{i}$, we still need to choose a particular point inside that tensor rule. That is what the local index α does.

The general formulas are

$$\xi^{\mathbf{i}, \alpha} = (\zeta_{i_1, \alpha_1}, \dots, \zeta_{i_b, \alpha_b}), \tag{24}$$

$$w^{\mathbf{i}, \alpha} = \prod_{j=1}^b \omega_{i_j, \alpha_j}. \tag{25}$$

In beginner language:

$\mathbf{i}$ chooses the rule level in each coordinate, while α chooses one actual node from inside those already-chosen rules.

8 What is sparse-grid level?

Now we come to the first of your two direct questions.

8.1 First plain-English answer

Sparse-grid level tells us:

which tensor rules are allowed to participate in the sparse-grid approximation.

It is a choice about the combination rule, not a choice about one particular coordinatewise tensor rule.

8.2 The formal active band

The teaching note uses the active band

$$\mathcal{A}_{b,L} = \{\mathbf{i} \in \mathbb{N}^b : L \leq |\mathbf{i}| \leq L + b - 1\}, \quad |\mathbf{i}| = i_1 + \cdots + i_b. \quad (26)$$

This says:

if the total index of a tensor label $\mathbf{i}$ lies between L and $L + b - 1$, then that tensor rule is allowed into the sparse-grid construction.

8.3 Two-dimensional example: sparse-grid level $L = 2$

Take

$$b = 2, \quad L = 2.$$

Then the active-band condition becomes

$$2 \leq |\mathbf{i}| \leq 3.$$

So the allowed tensor labels are exactly

$$(1, 1), \quad (1, 2), \quad (2, 1). \quad (27)$$

The tensor label $(2, 2)$ is not allowed, because

$$|(2, 2)| = 4, \quad 4 \notin [2, 3]. \quad (28)$$

So sparse-grid level 2 in two dimensions does not mean “use $(2, 2)$.” Instead it means “use the tensor rules whose total level lies in the allowed band for $L = 2$.”

8.4 Three-dimensional example: sparse-grid level $L = 2$

Take

$$b = 3, \quad L = 2.$$

Then the active-band condition becomes

$$2 \leq |\mathbf{i}| \leq 4.$$

Because each coordinate level is at least 1, the allowed tensor labels are

$$(1, 1, 1), \quad (1, 1, 2), \quad (1, 2, 1), \quad (2, 1, 1). \quad (29)$$

Again, the full tensor label $(2, 2, 2)$ is not allowed, because

$$|(2, 2, 2)| = 6, \quad 6 \notin [2, 4]. \quad (30)$$

Plain-language summary. Sparse-grid level does not tell us which one-dimensional rule is used in every coordinate. It tells us which *multi-coordinate tensor rules* are admitted into the sparse approximation.

9 What is Smolyak construction?

Now we come to the second direct question.

9.1 First plain-English answer

Smolyak construction means:

start with several tensor rules, attach plus or minus coefficients to them, add them together, and merge repeated points.

This is the key idea. Smolyak is not just “throw away some tensor points.” Smolyak is a signed combination of whole tensor rules.

9.2 The formal formula

The teaching note writes the sparse-grid rule as

$$A_{b,L}[F] = \sum_{\mathbf{i} \in \mathcal{A}_{b,L}} c_{\mathbf{i}} I_{\mathbf{i}}[F], \quad (31)$$

where the coefficients are

$$c_{\mathbf{i}} = (-1)^{L+b-1-|\mathbf{i}|} \binom{b-1}{L+b-1-|\mathbf{i}|}. \quad (32)$$

If that formula looks frightening, the beginner message is much simpler:

Smolyak tells us which tensor rules to combine and whether each one enters with plus sign or minus sign.

10 What does it mean that tensor rules are combined by Smolyak?

This is the question that usually needs the most concrete explanation.

10.1 First understand what one tensor rule is

A tensor rule such as $(1, 2)$ is already a list of points and weights. Another tensor rule such as $(2, 1)$ is another list of points and weights. The Smolyak construction does not choose one of them and discard the others. Instead it says:

add together the weighted sums from several tensor rules.

So when we say

$$A_{2,2}[F] = -I_{(1,1)}[F] + I_{(1,2)}[F] + I_{(2,1)}[F], \quad (33)$$

we mean:

1. evaluate F on the tensor rule $(1, 1)$,
2. evaluate F on the tensor rule $(1, 2)$,
3. evaluate F on the tensor rule $(2, 1)$,
4. subtract the first weighted sum,

5. add the second weighted sum,
6. add the third weighted sum.

That is what it means to say that the tensor rules are combined by Smolyak.

10.2 Concrete two-dimensional example, step by step

Take the one-dimensional rules

$$I_1 : (0, 1), \quad I_2 : \left\{ \left(-\sqrt{3}, \frac{1}{6} \right), \left(0, \frac{2}{3} \right), \left(\sqrt{3}, \frac{1}{6} \right) \right\}. \quad (34)$$

For sparse-grid level $L = 2$ in dimension $b = 2$, the active tensor rules are $(1, 1)$, $(1, 2)$, and $(2, 1)$. Their tensor points are:

$$(1, 1) : \{(0, 0)\}, \quad (35)$$

$$(1, 2) : \{(0, -\sqrt{3}), (0, 0), (0, \sqrt{3})\}, \quad (36)$$

$$(2, 1) : \{(-\sqrt{3}, 0), (0, 0), (\sqrt{3}, 0)\}. \quad (37)$$

The Smolyak coefficients are

$$c_{(1,1)} = -1, \quad c_{(1,2)} = 1, \quad c_{(2,1)} = 1. \quad (38)$$

So the combined rule is exactly

$$A_{2,2}[F] = -I_{(1,1)}[F] + I_{(1,2)}[F] + I_{(2,1)}[F]. \quad (39)$$

Now let us combine the point contributions one by one.

Contribution from $(1, 1)$. This rule contributes only the center point:

$$(0, 0) \quad \text{with signed weight} \quad -1.$$

Contribution from $(1, 2)$. This contributes

$$(0, -\sqrt{3}), (0, 0), (0, \sqrt{3})$$

with signed weights

$$\frac{1}{6}, \frac{2}{3}, \frac{1}{6}.$$

Contribution from $(2, 1)$. This contributes

$$(-\sqrt{3}, 0), (0, 0), (\sqrt{3}, 0)$$

with signed weights

$$\frac{1}{6}, \frac{2}{3}, \frac{1}{6}.$$

10.3 Now merge the repeated points

The point $(0,0)$ appears in all three tensor rules. So its final weight is obtained by adding its three signed contributions:

$$-1 + \frac{2}{3} + \frac{2}{3} = \frac{1}{3}. \quad (40)$$

The four axis points each appear only once, so they each keep weight $1/6$. Therefore the final merged sparse-grid cloud is

$$(0,0), \quad (\pm\sqrt{3},0), \quad (0,\pm\sqrt{3}), \quad (41)$$

with weights

$$\frac{1}{3}, \quad \frac{1}{6}, \quad \frac{1}{6}, \quad \frac{1}{6}, \quad \frac{1}{6}. \quad (42)$$

Why this cloud should remind you of a UKF picture. A beginner who has seen a UKF or Unscented Transform diagram should notice that this merged SGQF cloud looks familiar: there is one center point and several symmetric off-center points. That visual similarity is real and useful. Both constructions use a deterministic weighted point cloud to approximate Gaussian expectations. However, the final SGQF cloud was not chosen all at once by one sigma-point formula. It was built by combining the tensor rules $(1,1)$, $(1,2)$, and $(2,1)$ with Smolyak coefficients and then merging repeated points.

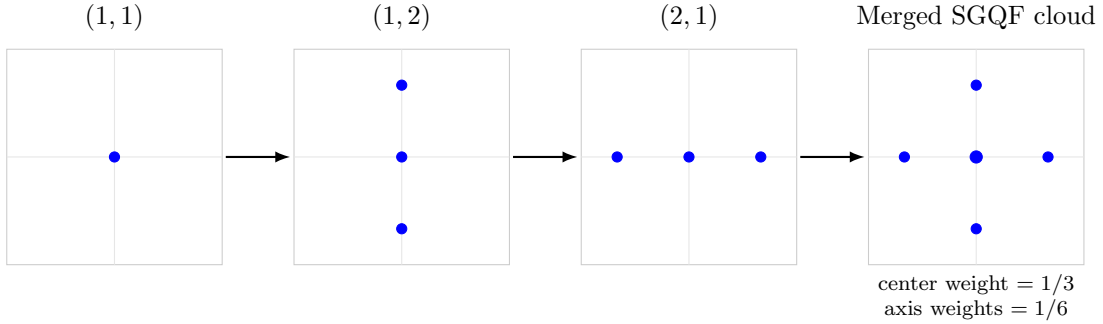


Figure 1: How the two-dimensional SGQF cloud is built: the active tensor rules $(1,1)$, $(1,2)$, and $(2,1)$ are combined, and repeated point contributions are merged. The final five-point cloud has the kind of central plus symmetric off-center geometry that may remind a reader of a UKF picture, but its construction rule is different.

A direct visual comparison with a UKF-style sketch. The next picture makes the similarity and difference explicit. The left panel is the merged SGQF cloud from Figure 1. The right panel is a canonical introductory UKF-style sigma-point sketch: one center point plus four symmetric axis points. The pictures look similar because both methods are deterministic weighted point clouds for Gaussian expectations. But the SGQF cloud comes from active tensor-rule selection and Smolyak recombination, whereas a UKF picture is usually presented as one directly specified sigma-point set.

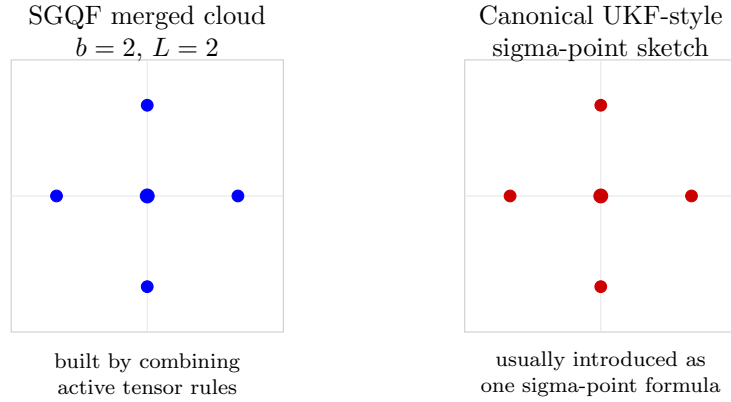


Figure 2: A safe visual analogy. The SGQF merged cloud and a canonical introductory UKF-style sigma-point sketch can look similar geometrically: both show a center point plus symmetric axis points. The similarity is useful for intuition, but the construction is different.

11 How sparse-grid rules compare with UKF-style sigma-point filters

The user asked specifically about UKF. That is the right comparison to make, because the pictures can look similar while the construction logic is different.

11.1 What is similar

At the most basic visual and computational level, SGQF clouds and UKF-style sigma-point sets have a real similarity:

- both are deterministic weighted point clouds,
- both are used to approximate Gaussian expectations,
- both are centered around the Gaussian mean,
- both often show symmetry around that center,
- both can be drawn as “center plus off-center points” diagrams.

So it is completely reasonable for a beginner to think, “this sparse-grid cloud looks a bit like a UKF picture.”

11.2 What is different in construction

This is the most important difference.

Typical beginner picture of UKF. A UKF is usually introduced as one directly specified sigma-point set: choose one formula for the sigma-point locations and one formula for the weights, then use that single point cloud to approximate Gaussian expectations.

SGQF construction. SGQF is built more indirectly. In the present note the construction is:

1. choose a one-dimensional rule family such as standard-normal GHQ,
2. choose a sparse-grid level L ,
3. determine which tensor rules are active,
4. attach Smolyak coefficients to those active tensor rules,
5. merge repeated point contributions.

So the final SGQF cloud is not usually chosen all at once by one sigma-point formula. It is assembled from several smaller tensor rules.

11.3 What the source paper says about UKF

The source-supported connection is narrower and more precise than “SGQF is just UKF.” Jia, Xin, and Cheng show that the unscented-transform point set appears as a level-2 sparse-grid member under the source’s one-point and three-point univariate choice. So the right beginner conclusion is:

UKF and low-level SGQF live in the same broad family of deterministic Gaussian point-set approximations, but SGQF is a more systematic sparse-grid construction, not merely a relabeling of the UKF.

11.4 When the final cloud can look similar to a UKF cloud

The user raised an important subtlety: if both SGQF and a UKF-style cloud are trying to match the same second moments, should the weights become the same or at least similar? The answer is: sometimes yes, but not always. Second-moment accuracy by itself does not uniquely determine the whole point cloud.

To see this, imagine a symmetric axis cloud in dimension d :

- one center point at the origin with weight w_0 ,
- two axis points $\pm ae_i$ in each coordinate direction,
- every off-center point has the same weight w .

There are therefore $2d$ off-center points.

If we want the weights to sum to one, we need

$$w_0 + 2dw = 1. \tag{43}$$

If we want the cloud to reproduce variance 1 in each coordinate, then only the pair $+ae_i$ and $-ae_i$ contributes to the second moment in that coordinate, so we need

$$2wa^2 = 1. \tag{44}$$

From (44) we get

$$w = \frac{1}{2a^2}, \quad w_0 = 1 - \frac{d}{a^2}. \tag{45}$$

So second-moment accuracy does not force one unique choice of weights. It only says that once the radius a is chosen, the weights are determined by that radius.

Two-dimensional example. If $d = 2$ and $a = \sqrt{3}$, then

$$w = \frac{1}{2 \cdot 3} = \frac{1}{6}, \quad w_0 = 1 - \frac{2}{3} = \frac{1}{3}. \quad (46)$$

So the resulting cloud has exactly the same weights as the merged two-dimensional SGQF level-2 cloud:

$$\text{center weight} = \frac{1}{3}, \quad \text{each axis weight} = \frac{1}{6}. \quad (47)$$

This is one reason the final SGQF picture can look very UKF-like.

Three-dimensional example. If $d = 3$ and again $a = \sqrt{3}$, then

$$w = \frac{1}{6}, \quad w_0 = 1 - \frac{3}{3} = 0. \quad (48)$$

So the center weight disappears and the six axis points each carry weight $1/6$. This matches the low-level three-dimensional SGQF toy cloud in the main note.

What this does and does not show. This calculation shows that if two methods end with the same symmetric axis geometry and both enforce the same second-moment conditions, then the final weights can indeed match. But it does *not* show that SGQF and UKF are the same construction. It only shows that the final low-level cloud can coincide in special cases.

11.5 Why sparse-grid level is not just “point spread”

A UKF is often introduced with parameters that directly control the final sigma-point geometry or weighting. By contrast, sparse-grid level L does not simply say “spread the same cloud farther out” or “keep the same pattern but change one knob.” Instead, sparse-grid level decides which tensor rules are allowed into the active band. The final cloud changes because the construction includes different tensor components, not because one predeclared sigma-point set was stretched.

11.6 Why Smolyak is not “delete a few points”

A beginner might imagine that SGQF starts from a full tensor grid and then merely deletes some inconvenient points. That is not what Smolyak does. Smolyak builds a signed combination of whole tensor rules. Some repeated points then merge, and some weights can be reduced or canceled by adding signed contributions together. So the final SGQF cloud is an algebraic recombination of tensor rules, not only a pruned subset of a larger grid.

11.7 The signed-weight difference

This is another useful contrast with the usual beginner UKF picture. In many introductory sigma-point diagrams, readers imagine the weights as simple positive masses attached to the points. Smolyak construction is more subtle: it combines tensor rules with signed coefficients. After merging, some final weights may be smaller than expected or even negative in other sparse-grid settings. So an SGQF cloud can resemble a sigma-point cloud geometrically while behaving differently algebraically.

11.8 Safe one-paragraph summary

A good beginner summary is this. A UKF-style sigma-point picture and a low-level SGQF cloud can look similar because both are deterministic weighted point sets used to approximate Gaussian expectations. But the UKF picture is usually taught as one directly specified sigma-point formula, whereas SGQF is built by choosing one-dimensional rules, selecting active tensor rules through sparse-grid level, and then combining those tensor rules by Smolyak coefficients. So the geometry may look familiar while the construction logic is different.

12 How sparse-grid level and Smolyak fit together

These two ideas fit together as follows.

12.1 Sparse-grid level chooses the active tensor rules

The sparse-grid level L tells us which tensor rules belong to the active band $\mathcal{A}_{b,L}$.

12.2 Smolyak tells us how to combine those active tensor rules

Once the active tensor rules have been selected, the Smolyak formula tells us what sign and coefficient each active tensor rule receives, and how their contributions are added together.

12.3 One-sentence summary

If you want one sentence to remember, use this one:

The sparse-grid level chooses which tensor rules are active, and the Smolyak construction tells us how to combine those active tensor rules into one sparse rule.

13 A full worked three-dimensional example using $F(\xi_1, \xi_2, \xi_3) = e^{\xi_1 \xi_2 \xi_3}$

Now let us put everything together in one explicit example.

13.1 Step 1: choose the function

Take

$$F(\xi_1, \xi_2, \xi_3) = e^{\xi_1 \xi_2 \xi_3}. \quad (49)$$

This means:

- if one coordinate is 0, then $F = 1$,
- if all three coordinates are nonzero, then F may be much larger or much smaller than 1.

13.2 Step 2: evaluate the simplest tensor rule $(1, 1, 1)$

The tensor rule $(1, 1, 1)$ uses I_1 in all three coordinates. So there is only one point:

$$(0, 0, 0).$$

Therefore

$$I_{(1,1,1)}[F] = F(0, 0, 0) = e^{0 \cdot 0 \cdot 0} = 1. \quad (50)$$

13.3 Step 3: evaluate the tensor rule $(2, 1, 1)$

The tensor rule $(2, 1, 1)$ uses I_2 in coordinate 1 and I_1 in coordinates 2 and 3. So the tensor points are

$$(-\sqrt{3}, 0, 0), \quad (0, 0, 0), \quad (\sqrt{3}, 0, 0).$$

At all three points we have $\xi_2 = 0$ and $\xi_3 = 0$, so

$$\xi_1 \xi_2 \xi_3 = 0, \quad F = e^0 = 1.$$

Therefore

$$I_{(2,1,1)}[F] = \frac{1}{6}(1) + \frac{2}{3}(1) + \frac{1}{6}(1) = 1. \quad (51)$$

By the same reasoning,

$$I_{(1,2,1)}[F] = 1, \quad I_{(1,1,2)}[F] = 1. \quad (52)$$

13.4 Step 4: choose sparse-grid level $L = 2$

Take dimension $b = 3$ and sparse-grid level $L = 2$. The active tensor rules are

$$(1, 1, 1), \quad (1, 1, 2), \quad (1, 2, 1), \quad (2, 1, 1). \quad (53)$$

The Smolyak coefficients are

$$c_{(1,1,1)} = -2, \quad c_{(1,1,2)} = 1, \quad c_{(1,2,1)} = 1, \quad c_{(2,1,1)} = 1. \quad (54)$$

Therefore

$$A_{3,2}[F] = -2I_{(1,1,1)}[F] + I_{(1,1,2)}[F] + I_{(1,2,1)}[F] + I_{(2,1,1)}[F]. \quad (55)$$

Using (50), (51), and (52), we get

$$A_{3,2}[F] = -2(1) + 1 + 1 + 1 = 1. \quad (56)$$

So in this example the sparse-grid level-2 rule gives the value 1.

13.5 Step 5: compare with the full tensor rule $(2, 2, 2)$

Now compare this with the full tensor rule $(2, 2, 2)$. This rule uses I_2 in all three coordinates, so it has

$$3 \times 3 \times 3 = 27 \quad (57)$$

raw tensor-product points.

Among those 27 points:

- 19 points have at least one coordinate equal to 0, so at those points $F = 1$,
- 8 corner points have all coordinates equal to $\pm\sqrt{3}$, so there $\xi_1 \xi_2 \xi_3 = \pm 3\sqrt{3}$ and $F = e^{\pm 3\sqrt{3}}$.

Each corner weight is

$$\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}. \quad (58)$$

There are 4 positive-sign corners and 4 negative-sign corners, so

$$I_{(2,2,2)}[F] = \frac{26}{27} + \frac{1}{54}e^{3\sqrt{3}} + \frac{1}{54}e^{-3\sqrt{3}}. \quad (59)$$

Numerically this is about

$$I_{(2,2,2)}[F] \approx 4.32, \quad (60)$$

which is very different from the sparse-grid level-2 value $A_{3,2}[F] = 1$.

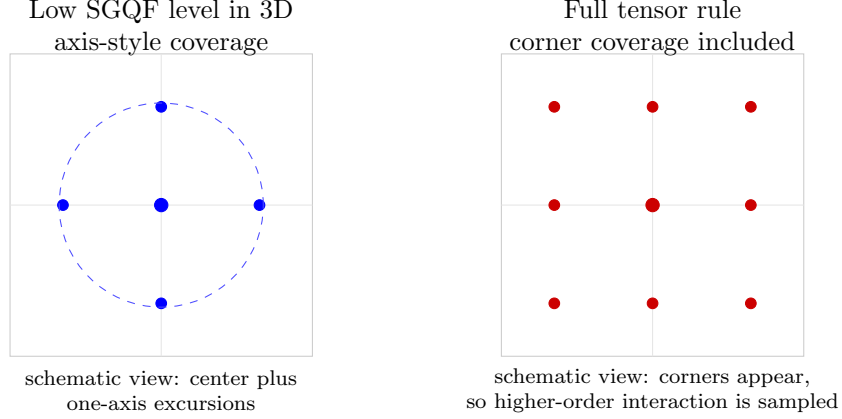


Figure 3: A schematic comparison for the three-variable example $F(\xi_1, \xi_2, \xi_3) = e^{\xi_1 \xi_2 \xi_3}$. A low sparse-grid level mainly sees center and axis-type structure, while the full tensor rule also includes corner coverage. For a function driven by a genuine three-way product, those corners matter.

13.6 What this example teaches

This example teaches something important and intuitive.

The function

$$F(\xi_1, \xi_2, \xi_3) = e^{\xi_1 \xi_2 \xi_3}$$

contains a genuine three-way interaction because all three coordinates appear in a product. The low sparse-grid level $L = 2$ in three dimensions mainly uses the center and one-axis rules. Those points all have at least one zero coordinate, so this low sparse-grid level cannot really see the full three-way corner interaction. By contrast, the full tensor rule $(2, 2, 2)$ does include the corners, so it does see those large values.

So this single example shows how all the pieces fit together:

- F is the function being integrated,
- I_1 and I_2 are one-dimensional GHQ rules,
- $(2, 1, 1)$ is a tensor label saying which 1D rule is used in each coordinate,
- sparse-grid level $L = 2$ chooses the active tensor rules,
- Smolyak tells us how to combine those active tensor rules,
- the final sparse-grid value can differ sharply from the full tensor value if the function depends strongly on interactions that the low sparse-grid level does not sample well.

14 What does the source notation N_q^b mean?

The source paper groups tensor labels by total index using

$$N_q^b = \{\mathbf{i} = (i_1, \dots, i_b) \in \mathbb{N}^b : i_1 + \dots + i_b = b + q\}. \quad (61)$$

This looks abstract, but it is only a grouping device.

14.1 Two-dimensional example

If $b = 2$, then

$$N_0^2 = \{(1, 1)\}, \quad (62)$$

$$N_1^2 = \{(1, 2), (2, 1)\}, \quad (63)$$

$$N_2^2 = \{(1, 3), (2, 2), (3, 1)\}. \quad (64)$$

So N_q^2 simply groups tensor labels according to their total sum.

14.2 Three-dimensional example

If $b = 3$, then

$$N_0^3 = \{(1, 1, 1)\}, \quad (65)$$

$$N_1^3 = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}. \quad (66)$$

Again, nothing mysterious is happening. The set N_q^b is just organizing the same tensor labels we already discussed.

Plain-language translation. When you see N_q^b , read it as:

“the set of all tensor labels in b dimensions whose total index is $b + q$.”

15 What the source paper means by exactness levels

The Jia–Xin–Cheng paper gives the following rule of thumb for the univariate family. A one-dimensional rule at level ℓ is chosen to be exact for univariate polynomials at least up to degree

$$2\ell - 1. \quad (67)$$

Then Theorem 3.1 says that the sparse-grid rule at level L is exact for multivariate polynomials of total degree up to

$$2L - 1. \quad (68)$$

So the safe beginner dictionary is:

- one-dimensional level 1 $\Rightarrow$ at least degree 1 exactness,
- one-dimensional level 2 $\Rightarrow$ at least degree 3 exactness,
- one-dimensional level 3 $\Rightarrow$ at least degree 5 exactness,
- sparse-grid level $L = 2 \Rightarrow$ multivariate total-degree exactness up to 3,
- sparse-grid level $L = 3 \Rightarrow$ multivariate total-degree exactness up to 5.

This explains why the number in the level label should not be read as a polynomial degree.

16 Did the original note define these things?

The fair answer is:

partly yes, but not in one place and not with the same level of beginner-friendly clarity for every concept.

More precisely:

- the original note defines the general one-dimensional rule I_ℓ ,
- it explicitly explains I_1 ,
- it explicitly derives the concrete rule I_2 ,
- it explains tensor labels such as $(1, 2)$,
- it later explains sparse-grid level L ,
- but it does not begin with one short subsection saying clearly that there are different uses of the word “level,” and it does not explain Smolyak combination in one slow, explicit, first-year-undergraduate style.

That is why a new reader can honestly feel that the notation is being used before its meaning has settled.

17 Common misreadings and the corrected reading

Misreading	Correct reading
“Level-2 rule means degree-2 polynomial.”	No. It means the second member of the one-dimensional rule family.
“Level 3 means degree 3.”	No. In the source-paper convention, level 3 means at least fifth-degree univariate exactness.
“(2, 2) is the same thing as sparse-grid level 2.”	No. (2, 2) is a tensor label; $L = 2$ is a sparse-grid combination level.
“Sparse-grid level $L = 2$ keeps the full tensor rule (2, 2) or (2, 2, 2).”	No. It combines lower-total-index tensor rules according to the active band.
“Smolyak just means delete a few points from the full tensor grid.”	No. Smolyak means combine whole tensor rules with signed coefficients and merge repeated points.
“The index α chooses the rule level.”	No. The level is chosen by $\mathbf{i}$; α chooses a point inside the already-chosen tensor rule.

18 A final memory aid

If you want one memory aid that fits on a small card, use this one:

I_2 : one-dimensional standard-normal GHQ rule,
 $(2, 2)$: use I_2 in both coordinates,

$L = 2$: choose which tensor rules are active,
 Smolyak : tell us how to combine those active tensor rules,
 F : the function being integrated,
 N_q^b : group tensor labels by their total index.

If you keep those six lines separate in your head, the notation becomes much less slippery.

References

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